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LAB Assignment (Studio 3T)

COLLECTION 1: flights

Schema

```
{  
  flight_id: Number,  
  airline: String,  
  source: String,  
  destination: String,  
  price: Number,  
  duration: Number  
}
```

COLLECTION 2: bookings

Schema

```
{  
  booking_id: Number,  
  passenger_name: String,  
  flight_id: Number,  
  city: String,  
  seats: Number,  
  total_fare: Number  
}
```

MongoDB query INSERTION

```
use airlineDB
```

```
db.createCollection("flights")  
db.createCollection("bookings")
```

```
db.flights.insertMany([  
  { flight_id: 1, airline: "Indigo", source: "Delhi", destination: "Mumbai", price: 5000, duration: 2 },  
  { flight_id: 2, airline: "AirIndia", source: "Kolkata", destination: "Chennai", price: 6500, duration: 3 },  
  { flight_id: 3, airline: "SpiceJet", source: "Mumbai", destination: "Bangalore", price: 4000, duration: 1.5 },  
  { flight_id: 4, airline: "Vistara", source: "Delhi", destination: "Chennai", price: 7000, duration: 3 },  
  { flight_id: 5, airline: "GoFirst", source: "Mumbai", destination: "Delhi", price: 4800, duration: 2 },  
  { flight_id: 6, airline: "Indigo", source: "Bangalore", destination: "Kolkata", price: 5500, duration: 2.5 }])
```

```

db.bookings.insertMany([
  { booking_id: 101, passenger_name: "Aarav", flight_id: 1, city: "Delhi", seats:
1, total_fare: 5000 },
  { booking_id: 102, passenger_name: "Meera", flight_id: 2, city: "Kolkata", seats:
2, total_fare: 13000 },
  { booking_id: 103, passenger_name: "Rohan", flight_id: 3, city: "Mumbai", seats:
1, total_fare: 4000 },
  { booking_id: 104, passenger_name: "Sneha", flight_id: 4, city: "Delhi", seats:
3, total_fare: 21000 },
  { booking_id: 105, passenger_name: "Kunal", flight_id: 5, city: "Mumbai", seats:
2, total_fare: 9600 },
  { booking_id: 106, passenger_name: "Priya", flight_id: 6, city: "Bangalore",
seats: 1, total_fare: 5500 }
])

```

db.bookings.find()

bookings > booking_id					
_id	booking_id	passenger_name	flight_id	city	seats
69ca1a2a817c13...	101	Aarav	1	Delhi	1
69ca1a2a817c13...	102	Meera	2	Kolkata	2
69ca1a2a817c13...	103	Rohan	3	Mumbai	1
69ca1a2a817c13...	105	Kunal	5	Mumbai	2
69ca1a2a817c13...	106	Priya	6	Bangalore	1

QUESTIONS

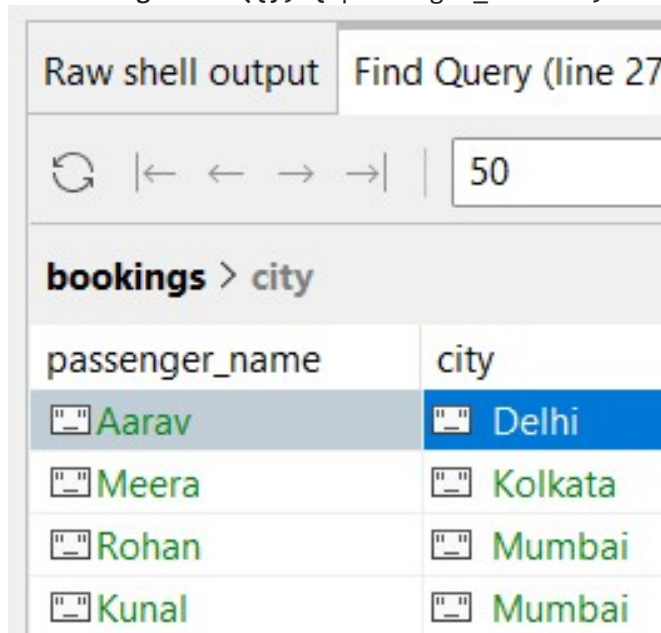
Q1. Display all flights.

db.flights.find()

flights > flight_id					
_id	flight_id	airline	source	destination	price
69ca1a23817c13...	1	Indigo	Delhi	Mumbai	5500
69ca1a23817c13...	2	AirIndia	Kolkata	Chennai	6500
69ca1a23817c13...	3	SpiceJet	Mumbai	Bangalore	4000
69ca1a23817c13...	4	Vistara	Delhi	Chennai	7000
69ca1a23817c13...	5	GoFirst	Mumbai	Delhi	4000

Q2. Display passenger_name and city from bookings.

```
db.bookings.find({}, { passenger_name: 1, city: 1, _id: 0 })
```

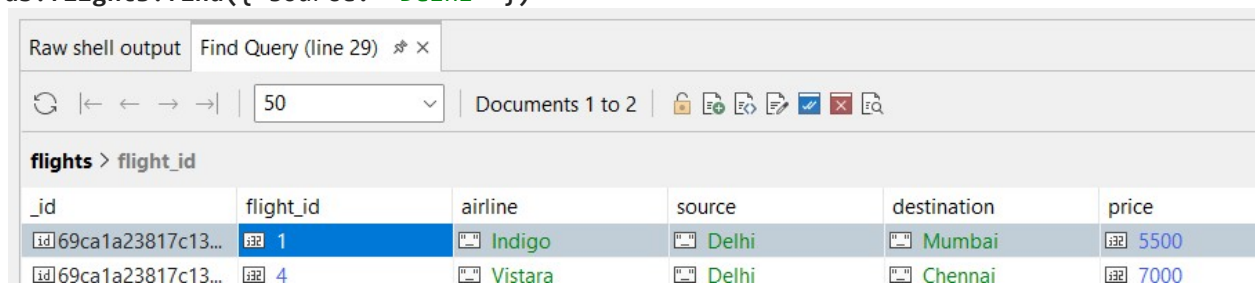


The screenshot shows the MongoDB Shell interface. The top bar has tabs for 'Raw shell output' and 'Find Query (line 27)'. Below the tabs is a navigation bar with a refresh icon, left and right arrows, and a page size dropdown set to 50. The main content area displays the command 'bookings > city' followed by a table of results. The table has two columns: 'passenger_name' and 'city'. The data rows are: Aarav (Delhi), Meera (Kolkata), Rohan (Mumbai), and Kunal (Mumbai). Each cell contains a small document icon.

passenger_name	city
Aarav	Delhi
Meera	Kolkata
Rohan	Mumbai
Kunal	Mumbai

Q3. Find flights where source is Delhi.

```
db.flights.find({ source: "Delhi" })
```



The screenshot shows the MongoDB Shell interface. The top bar has tabs for 'Raw shell output' and 'Find Query (line 29)'. Below the tabs is a navigation bar with a refresh icon, left and right arrows, a page size dropdown set to 50, and a status bar showing 'Documents 1 to 2'. The main content area displays the command 'flights > flight_id' followed by a table of results. The table has six columns: '_id', 'flight_id', 'airline', 'source', 'destination', and 'price'. The data rows are: flight_id 1 (Indigo, Delhi to Mumbai, price 5500) and flight_id 4 (Vistara, Delhi to Chennai, price 7000). Each cell contains a small document icon.

_id	flight_id	airline	source	destination	price
69ca1a23817c13...	1	Indigo	Delhi	Mumbai	5500
69ca1a23817c13...	4	Vistara	Delhi	Chennai	7000

Q4. Find bookings where seats = 1.

```
db.bookings.find({ seats: 1 })
```

Raw shell outputFind Query (line 31) ✖

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Documents 1 to 3

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bookings > booking_id

_id	booking_id	passenger_name	flight_id	city	seats
[id]69ca1a2a817c13...	101	Aarav	1	Delhi	1
[id]69ca1a2a817c13...	103	Rohan	3	Mumbai	1

Q5. Sort flights by price in ascending order.

```
db.flights.find().sort({ price: 1 })
```

Raw shell output

Find Query (line 33) ✖

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Q6. Display top 2 bookings with highest total_fare.

```
db.bookings.find().sort({ total_fare: -1 }).limit(2)
```

bookings > booking_id

_id	booking_id	passenger_name	flight_id	city	seats
[id]69ca1a2a817c13...	102	Meera	2	Kolkata	2
[id]69ca1a2a817c13...	105	Kunal	5	Mumbai	2

Q7. Find flights where price is greater than 5000.

```
db.flights.find({ price: { $gt: 5000 } })
```

flights > flight_id

_id	flight_id	airline	source	destination	price
[id]69ca1a23817c13...	1	Indigo	Delhi	Mumbai	5500
[id]69ca1a23817c13...	2	AirIndia	Kolkata	Chennai	6500
[id]69ca1a23817c13...	4	Vistara	Delhi	Chennai	7000

Q8. Display flights where destination is not Chennai.

```
db.flights.find({ destination: { $ne: "Chennai" } })
```

flights > flight_id					
_id	flight_id	airline	source	destination	price
id 69ca1a23817c13...	1	Indigo	Delhi	Mumbai	5500
id 69ca1a23817c13...	3	SpiceJet	Mumbai	Bangalore	4000
id 69ca1a23817c13...	5	GoFirst	Mumbai	Delhi	4800

Q9. Find bookings where seats are between 1 and 2.

```
db.bookings.find({ seats: { $gte: 1, $lte: 2 } })
```

bookings > booking_id					
_id	booking_id	passenger_name	flight_id	city	seats
id 69ca1a2a817c13...	101	Aarav	1	Delhi	1
id 69ca1a2a817c13...	102	Meera	2	Kolkata	2
id 69ca1a2a817c13...	103	Rohan	3	Mumbai	1
id 69ca1a2a817c13...	105	Kunal	5	Mumbai	2

Q10. Update price by +500 for flights of airline "Indigo".

```
db.flights.updateMany(
  { airline: "Indigo" },
  { $inc: { price: 500 } }
)
```

Raw shell output Shell Output (Documents) ✖				
< < > > 50 Documents 1 to 1				
Result > insertedId				
acknowledged	insertedId	matchedCount	modifiedCount	ups

Q11. Delete bookings where seats are greater than 2.

```
db.bookings.deleteMany({ seats: { $gt: 2 } })
```

Result > deletedCount	
acknowledged	deletedCount
☒ true	0.0

Q12. Find total fare collected city-wise. (Aggregation)

```
db.bookings.aggregate([ { $group: { _id: "$city", total_fare_collected: { $sum: "$total_fare" } } } ])
```

bookings > total_fare_collected	
_id	total_fare_collected
Kolkata	13000
Delhi	5000
Bangalore	5500
Mumbai	13500

Q13. Find average flight price per airline. (Aggregation)

```
db.flights.aggregate([ { $group: { _id: "$airline", Avg_flight_price: { $avg: "$price" } } } ])
```

flights > Avg_flight_price	
_id	Avg_flight_price
Indigo	6250.0
Vistara	7000.0
GoFirst	4800.0
AirIndia	6500.0

Q14. Display airlines where average price is greater than 5500. (Aggregation + filter)

```
db.flights.aggregate([ { $group: { _id: "$airline", avg_price: { $avg: "$price" } } },  
 { $match: { avg_price: { $gt: 5500 } } } ])
```

flights > avg_price	
_id	avg_price
"Indigo"	6250.0
"Vistara"	7000.0
"AirIndia"	6500.0

Q15. Find maximum total_fare among bookings for each city. (Aggregation)

```
db.bookings.aggregate([ { $group: { _id: "$city", max_total_fare: { $max:  
"$total_fare" } } } ])
```

bookings > max_total_fare	
_id	max_total_fare
"Mumbai"	9600
"Bangalore"	5500